

Remark 3. According to Remark 2, each I_{ij} can be written as an interval $I_{ij} = [L_{ij}, U_{ij}]$ where L_{ij} and U_{ij} are the lower and upper bounds of the interval, respectively. Also, each S_{ij} (if it is non-empty) is obtained as one of the forms of $S_{ij} = \{L_{ij}\}$, $S_{ij} = \{U_{ij}\}$, $S_{ij} = \{L_{ij}, U_{ij}\}$ or $S_{ij} = [L_{ij}, U_{ij}]$.

Definition 4. For each $j \in \mathcal{J}$, we define $I_j = \bigcap_{i \in \mathcal{I}} I_{ij}$. Also, let $S'_{ij} = S_{ij} \cap I_j$, $\forall i \in \mathcal{I}$ and $\forall j \in \mathcal{J}$.

Corollary 4. Let $i \in \mathcal{I}$ and $j \in \mathcal{J}$. If $S'_{ij} \neq \emptyset$, then S'_{ij} is as one of the forms of $S'_{ij} = \{L_j\}$, $S'_{ij} = \{U_j\}$, $S'_{ij} = \{L_j, U_j\}$ or $S'_{ij} = [L_j, U_j]$, where L_j and U_j are the lower and upper bound of $I_j = [L_j, U_j]$, respectively.

Proof. From Definition 4 and Remark 3, we have

$$I_j = \bigcap_{i \in \mathcal{I}} I_{ij} = \bigcap_{i \in \mathcal{I}} [L_{ij}, U_{ij}] = \left[\max_{i \in \mathcal{I}} \{L_{ij}\}, \min_{i \in \mathcal{I}} \{U_{ij}\} \right] = [L_j, U_j]$$

Hence, for each $i \in \mathcal{I}$, if $L_{ij} \in I_j$ ($U_{ij} \in I_j$), then we must have $L_{ij} = L_j$ ($U_{ij} = U_j$). On the other hand, from Remark 3, a non-empty S_{ij} is as one of the forms of $S_{ij} = \{L_{ij}\}$, $S_{ij} = \{U_{ij}\}$, $S_{ij} = \{L_{ij}, U_{ij}\}$ or $S_{ij} = [L_{ij}, U_{ij}]$. So, if $S_{ij} = \{L_{ij}\}$ and $S'_{ij} = S_{ij} \cap I_j \neq \emptyset$, then $L_{ij} \in I_j$; that is, $L_{ij} = L_j$ and therefore $S'_{ij} = S_{ij} \cap I_j = \{L_j\}$. For the other cases, the results are attained by the similar argument. $\square$

Example 1. Consider Problem (1) includes Yager operator where

$$z = c^T x = x_1 + 0.35x_2 + 0.93x_3 + 3.28x_4 + 5.03x_5 + 2.96x_6 + x_7 + 2.75x_8 + 5.25x_9 + 6.39x_{10}$$

$$A^+ = \begin{bmatrix} 0.25 & 0.32 & 0.41 & 0.19 & 0.70 & 0.13 & 0.44 & 0.37 & 0.28 & 0.50 \\ 0.80 & 0.73 & 0.64 & 0.79 & 0.80 & 0.22 & 0.80 & 0.56 & 0.10 & 0.28 \\ 0.11 & 0.20 & 0.12 & 0.13 & 0.05 & 0.25 & 0.40 & 0.25 & 0.20 & 0.18 \\ 0.10 & 0.23 & 0.25 & 0.15 & 0.12 & 0.05 & 0.02 & 0.01 & 0.15 & 0.15 \\ 0.45 & 0.35 & 0.70 & 0.50 & 0.41 & 0.27 & 0.39 & 0.48 & 0.17 & 0.39 \\ 0.60 & 0.70 & 0.25 & 0.38 & 0.63 & 0.58 & 0.46 & 0.47 & 0.85 & 0.33 \\ 0.01 & 0.02 & 0.15 & 0.09 & 0.12 & 0.10 & 0.15 & 0.15 & 0.04 & 0.25 \\ 0.75 & 0.64 & 0.32 & 0.29 & 0.39 & 0.61 & 0.57 & 0.34 & 1.00 & 0.46 \\ 0.22 & 0.20 & 0.35 & 0.23 & 0.30 & 0.18 & 0.29 & 0.25 & 0.35 & 0.10 \\ 0.41 & 0.25 & 0.50 & 0.20 & 0.56 & 0.60 & 0.59 & 0.60 & 0.47 & 0.31 \end{bmatrix}$$

$$A^- = \begin{bmatrix} 0.70 & 0.70 & 0.32 & 0.44 & 0.00 & 0.16 & 0.20 & 0.50 & 0.40 & 0.39 \\ 0.70 & 0.65 & 0.14 & 0.12 & 0.80 & 0.76 & 0.00 & 1.00 & 0.15 & 0.79 \\ 0.17 & 0.24 & 0.20 & 0.20 & 0.06 & 0.25 & 0.13 & 0.19 & 0.22 & 0.02 \\ 0.14 & 0.10 & 0.04 & 0.00 & 0.10 & 0.00 & 0.14 & 0.02 & 0.15 & 0.08 \\ 0.70 & 0.04 & 0.27 & 0.36 & 0.60 & 0.40 & 0.48 & 0.50 & 0.50 & 0.50 \\ 0.66 & 0.63 & 0.14 & 0.73 & 0.53 & 0.46 & 0.61 & 0.85 & 0.85 & 0.39 \\ 0.00 & 0.15 & 0.15 & 0.05 & 0.02 & 0.03 & 0.10 & 0.12 & 0.18 & 0.09 \\ 0.63 & 0.03 & 0.55 & 0.77 & 0.79 & 0.49 & 0.21 & 0.32 & 0.80 & 0.71 \\ 0.27 & 0.30 & 0.35 & 0.24 & 0.35 & 0.07 & 0.29 & 0.35 & 0.20 & 0.75 \\ 0.59 & 0.34 & 0.26 & 0.38 & 0.02 & 0.60 & 0.52 & 0.43 & 0.27 & 0.44 \end{bmatrix}$$

$$b^T = [0.50, 0.80, 0.25, 0.15, 0.50, 0.75, 0.15, 0.80, 0.35, 0.60]$$

In this example, $\mathcal{I} = \{1, 2, \dots, 10\}$, $\mathcal{J} = \{1, 2, \dots, 10\}$ and φ is the Yager t-norm (that is continuous and nilpotent) with $p = 2$. So, according to Table A1 we have

$$\varphi(x, y) = T_Y^2(x, y) = \max \left\{ 0, 1 - ((1-x)^2 + (1-y)^2)^{1/2} \right\}$$